

HOPITAL

• $f, g: I \setminus \{x_0\} \rightarrow \mathbb{R}$, I INTERVALLO, $x_0 \in I$

ASSUMIAMO:

a) f, g SONO DERIVABILI E

$$g'(x) \neq 0 \quad \forall x \in I \setminus \{x_0\}$$

b) ABBIAMO UNA f i:

$$\lim_{x \rightarrow x_0} f(x) = 0$$

OPPURE

$$\lim_{x \rightarrow x_0} f(x) = \pm \infty$$

$$\lim_{x \rightarrow x_0} g(x) = 0$$

$$\lim_{x \rightarrow x_0} g(x) = \pm \infty$$

$$\left[\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = ? \right]$$

$$c) \exists \lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)} = l \in \mathbb{R} \cup \{\pm \infty\}$$

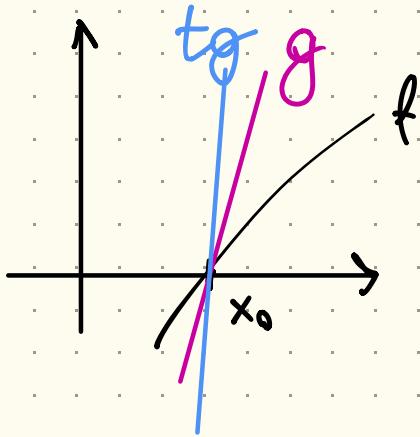
• ALLORA

$$\exists \lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = l$$

! NON BISOGNA FARE

$$\lim_{x \rightarrow x_0} \left(\frac{f(x)}{g(x)} \right)'$$

• OSSERVAZIONE $\Rightarrow$ PERCHÉ É VERO?



$$\frac{f(x)}{g(x)} \approx \frac{\overset{0=}{\cancel{f(x_0)}} + \cancel{f'(x_0)(x-x_0)}}{\overset{0=}{\cancel{g(x_0)}} + \cancel{g'(x_0)(x-x_0)}}$$

• ESEMPI

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{(\sin x)'}{x'} = \lim_{x \rightarrow 0} \frac{\cos x}{1} = \cos 0 = 1$$

$\nearrow \sin 0 = 0$
 $\searrow 0$

$$\lim_{x \rightarrow \pi} \frac{\sin x}{x} = \frac{0}{\pi} = 0$$

$\nearrow \sin \pi = 0$
 $\searrow \pi$

NO HOPITAL

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} \stackrel{H}{=} \frac{(1 - \cos x)'}{(x^2)'} = \lim_{x \rightarrow 0} \frac{\sin x}{2x} = \frac{1}{2}$$

$\nearrow 1 - \cos 0 = 0$
 $\searrow 0$
 $\nearrow 1$

es) DERIVO SOLO LE FORME INDETERMINATE

$$\bullet \lim_{x \rightarrow -1} \frac{x^5 - 2x^4 - x + 2}{x^2 + 3x^6 + 2}$$

$$\frac{f(x)}{g(x)} \quad \begin{array}{l} f(-1) = 0 \\ g(-1) = 0 \end{array}$$

$$\stackrel{H}{=} \lim_{x \rightarrow -1} \frac{5x^4 - 8x^3 - 1}{7x^6 + 18x^5} = -\frac{12}{11}$$

DERIVO

$$\bullet \lim_{x \rightarrow 0} \frac{\log \cos x}{x^2} = \log \cos 0 = \log 1 = 0$$

$$\stackrel{H}{=} \lim_{x \rightarrow 0} \frac{(\log \cos x)'}{(x^2)'} = \lim_{x \rightarrow 0} \frac{\log'(\cos x) \cdot (\cos x)'}{(x^2)'} =$$

$$= \lim_{x \rightarrow 0} \frac{\log'(\cos x) \cdot (\cos x)'}{2x}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{\cos x} \cdot (-\sin x)}{2x}$$

$$= \lim_{x \rightarrow 0} \left(\frac{1}{2 \cos x} \right) \cdot \left(\frac{\sin x}{x} \right) = -\frac{1}{2}$$

$\Rightarrow -\frac{1}{2} \cdot 1 = -\frac{1}{2}$

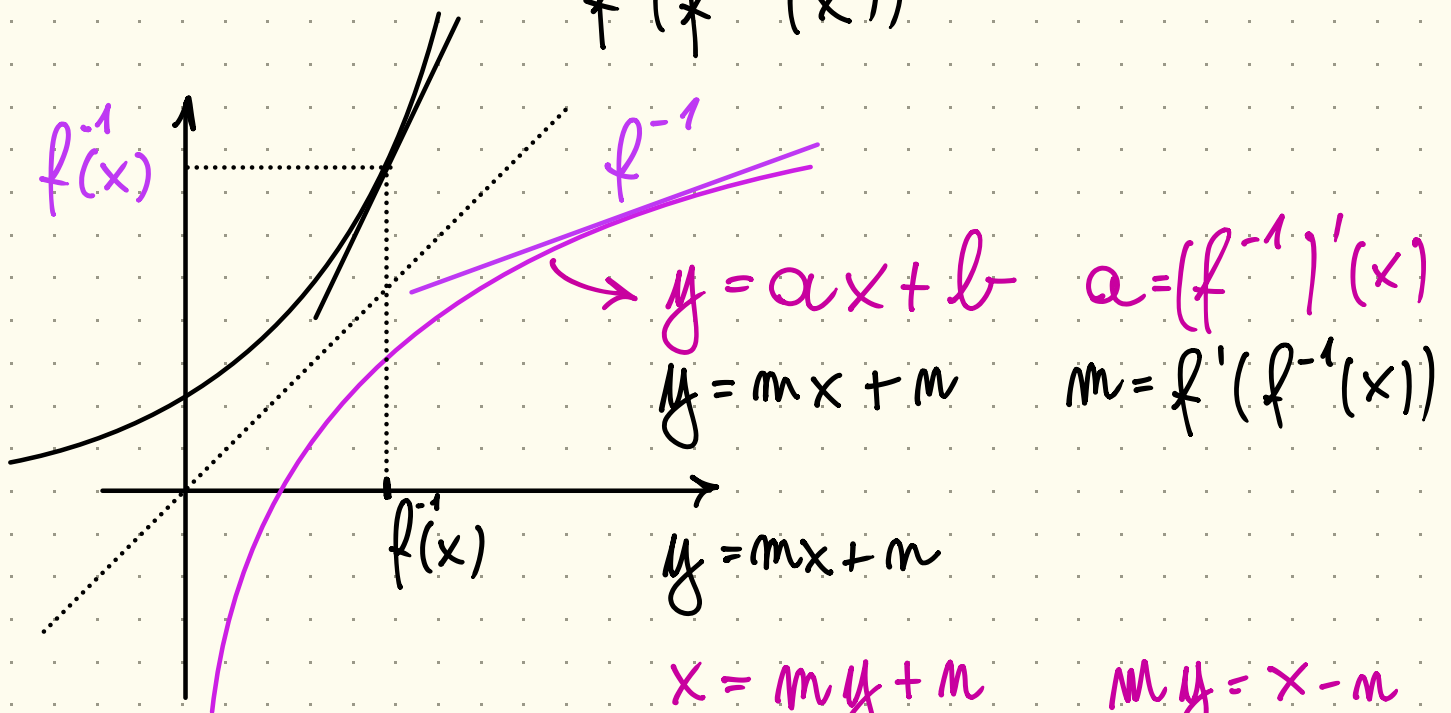
• TEOREMA $\rightarrow$ CONTINUITÀ E DERIVABILITÀ DELLA FUNZIONE INVERSA

• SIA $f: I \rightarrow \mathbb{R}$, I INTERVALLO, CONTINUA E INIE.
 ALLORA $\text{Im } f$ È UN INTERVALLO E f^{-1} È CONTINUA

INOLTRE SE f È DERIVABILE E $f'(x) \neq 0$
 $\forall x \in I$, ALLORA f^{-1} È DERIVABILE E

$\Downarrow$

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))} \quad \forall x \in I$$



es) DERIVATA DELL'INVERSA

$$\begin{aligned} \cdot (f^{-1})'(x) &= \frac{1}{f'(f^{-1}(x))} = \frac{1}{3(f^{-1}(x))^2} = \\ &= \frac{1}{3(\sqrt[3]{x})^2} = \frac{1}{3x^{\frac{2}{3}}} = \frac{1}{3} x^{-\frac{2}{3}} \quad \forall x \neq 0 \\ f^{-1} &\text{ NON É DERIVABILE IN } 0 \end{aligned}$$

- LA TANGENTE ORIZZONTALE VIENE TRASFORMATO IN UNA TANGENTE VERTICALE

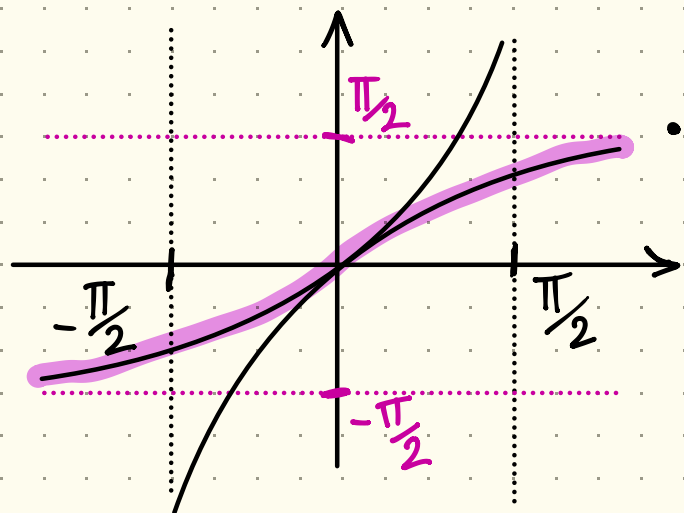
es) $f(x) = e^x$
 $f^{-1}(x) = \log x$
 $f'(x) = e^x$

$$\begin{aligned} (\log x)' &= \frac{1}{f'(f^{-1}(x))} = \frac{1}{f'(\log x)} = \frac{1}{e^{\log x}} = \\ &= \frac{1}{x} \end{aligned}$$

$$\bullet f(x) = \tan x \quad x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$f'(x) = 1 + \tan^2 x$$

$$f^{-1}(x) = \arctan x$$



$$\bullet (f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$

$$= \frac{1}{f'(\arctan x)}$$

$$= \frac{1}{1 + \tan^2 \arctan x}$$

↓
x²

$$= \frac{1}{1 + x^2}$$

es) $f(x) = \sin x, x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

$$f^{-1}(x) = \arcsin x$$

$$f'(x) = \cos x$$

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))} = \frac{1}{\cos(\arcsin x)} =$$

$$\overset{=}{=} \frac{1}{\sqrt{1 - \sin^2(\arcsin)}} = \frac{1}{\sqrt{1 - x^2}}$$

$$\cos x = \sqrt{1 - \sin^2 x} \quad \forall x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

QUINDI

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